

Network Systems and Security

Assignment: Network Traffic Analysis

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You have been appointed as a **Network and System Security Analyst** in your organisation. As part of your role, you are provided with network traffic logs collected from a firewall or an Intrusion Detection System (IDS). These logs capture detailed information about network flows, including communication between IP addresses, packet-level statistics, protocol usage, and timestamps.

To streamline the analysis process and avoid unnecessary preprocessing overhead, the data is provided in **CSV format** instead of raw XML. This allows you to focus directly on extracting insights rather than handling data format conversions.

Your objective is to analyse this data to uncover meaningful patterns, identify anomalies, and assess potential security threats. Since real-world network monitoring systems operate under memory constraints, it is often impractical to store complete information for all traffic flows. Therefore, certain tasks in this assignment require you to design and use **space-efficient (sublinear) algorithms** instead of maintaining full datasets.

You are encouraged to make reasonable design decisions, including choosing appropriate thresholds, time windows, and other parameters, based on your analysis.

1. Basic Network Traffic Statistics

To begin, you are required to parse the provided CSV files and extract key characteristics of the network traffic. Perform the following analyses:

1. Compute the total number of network flows in the dataset.
2. Identify the top k most frequently used protocols, where k is a configurable parameter (default: 5).
3. Identify the top 10 most active source IPs and destination IPs based on traffic volume. Clearly specify the metric used (e.g., number of flows, packets, or bytes).
4. Compute the average packet size across all flows. Additionally, comment on whether this average is representative of the dataset.
5. Identify the top 3 most frequent source–destination pairs.
6. Determine which IP addresses exhibit consistent communication patterns across multiple time windows. Clearly define your notion of a “*time window*” and justify your choice.
7. Detect and highlight irregular spikes in traffic volume over time. Specify the method used to define a “*spike*” and discuss possible causes.

8. Compute the variance of packet sizes. Analyse and explain potential reasons for high or low variance in the context of network behaviour.

2. Traffic Estimation Using Sublinear Space

In large-scale network monitoring systems, storing complete information for every IP address and packet count is often infeasible due to memory constraints. In this section, you are required to design and implement space-efficient (sublinear) techniques for traffic estimation.

(a) Estimating the Number of Unique IP Addresses

1. Design an approach to estimate the total number of distinct IP addresses using **sublinear space**, without explicitly storing all unique IPs.
2. Compute the same metric using a full linear-space approach, and compare the results with your sublinear estimate.
3. Ensure that the error rate (difference between sublinear and exact results) does not exceed **10%**.
4. Analyse the trade-offs between space usage, accuracy, and computation time.

(b) Identifying Frequently Contacted Destination IPs

1. Identify **heavy hitters**, i.e., destination IP addresses that receive a significant fraction of the total traffic.
2. Develop a space-efficient method to approximate the most frequently contacted destination IPs without maintaining exact per-IP counts.
3. Compare the approximate results with exact counts obtained using full memory, and evaluate the accuracy of your method.
4. Ensure that the error rate remains **below 10%**.

(c) Efficient Membership Testing for Blocklists

1. Design a method to determine whether an IP address has been previously observed, *without* storing all IPs explicitly.
2. Implement a **Bloom filter**-based solution for approximate membership testing.
3. Compare the false positive rate and associated trade-offs with an exact membership test that uses full space.

Hints: Explore techniques such as Bloom filters or any other sublinear technique.

3. Advanced Anomaly Detection

Anomaly detection in network traffic is essential for identifying potential security threats. In this section, you are required to analyse the data and detect various types of anomalous behaviour.

(a) Statistical Traffic Analysis

1. Define appropriate statistical thresholds to identify anomalies in packet sizes, flow counts, and protocol distributions.
2. Compare traffic patterns across different time windows (e.g., hourly vs. daily) to identify significant deviations.
3. Identify and flag IP addresses that generate unusually high or low volumes of traffic compared to expected behaviour.

(b) Behavioural Analysis

1. Identify IP addresses that exhibit significant changes in behaviour over time (e.g., sudden spikes in activity following periods of inactivity).
2. Detect network-wide correlation patterns, such as multiple IPs initiating communication with the same target within a short time interval.
3. Identify destination IPs that receive traffic from a large number of distinct sources within a short duration.

(c) Suspicious Communication Patterns

1. Identify long-duration connections that remain active significantly longer than typical flows.
2. Detect IPs that communicate using multiple protocols within a short time span.

4. Deep Security Threat Analysis

Building upon the anomalies identified earlier, you are now required to perform a deeper threat-hunting analysis using the network traffic data.

(a) Detecting Complex Attack Patterns

1. Identify **stealthy port scan** activities where an attacker probes ports over an extended period to avoid detection.
2. Detect **slow Distributed Denial-of-Service (DDoS)** attacks that may not exhibit sharp traffic spikes but gradually consume system resources.
3. Identify **IP hopping** behaviour, where a single attacker uses multiple IP addresses to evade detection mechanisms.

(b) Malicious Payload Identification

1. Analyse payload data within network flows to identify unusual or suspicious patterns.
2. Flag encrypted traffic that deviates from typical encrypted communication behaviour.
3. Identify network flows that resemble known **malware command-and-control (C2)** communication patterns.

(c) Threat Attribution and Risk Analysis

1. Categorise each detected security event into one of the following risk levels:
 - (i) Low risk
 - (ii) Medium risk
 - (iii) High riskProvide justification for each classification based on observed behaviour.
2. Generate a concise **risk report** summarising your findings.

Submission Guidelines

- Submit a **Jupyter Notebook** or **Python script** with your analysis.
- Include detailed comments and explanations in your code.
- Provide a short **PDF report** summarising your findings.